

HALF-YEARLY EXAMINATION 2025-26

CLASS XI - Artificial Intelligence (Set 2)

Maximum Marks: 70

Time Allowed: 3 Hour

Instructions:

- Attempt all questions from Section A and B and **any four questions** from Section C.
 - The intended marks for questions or parts of questions are given in brackets [].
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Section A: (Attempt all questions)

Question 1: Choose the correct answer from the options given below. [1 × 10 = 10]

- I. Which of the following is an application of Artificial Intelligence in healthcare?
a) Crop rotation b) Patient monitoring c) Bridge construction d) Traffic signal design
- II. Which of the following is a real-world application of Machine Learning?
a) Image recognition b) Mathematical calculation c) 3D drafting d) Hand crafting
- III. Which of these is an example of a voice-based assistant?
a) Paint b) Excel c) Siri d) Blender
- IV. Google Translate is an example of:
a) Voice modulation b) Language translation c) Data mining d) Image recognition
- V. The transpose of a matrix is obtained by:
a) Interchanging rows and columns b) Rotating the matrix c) Dividing each element by 2
d) Adding rows
- VI. Which logic gate gives output ‘1’ when at least one input is true?
a) AND b) OR c) XOR d) NOR
- VII. Which method in Python is used to add a element at the end of an existing list?
a) append() b) insert() c) extend() d) add()
- VIII. What will “Python”[2:6]‘ output?
a) Pyth b) ytho c) thon d) yth
- IX. In which learning approach is the data **not** labeled?
a) Supervised b) Unsupervised c) Reinforcement d) Deep Learning
- X. If $\vec{a} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, then $\vec{a} + \vec{b} = ?$
a) $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$ b) $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ c) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ d) $\begin{bmatrix} 5 \\ 12 \end{bmatrix}$

Section B: (Attempt all questions)

Question 2: Answer briefly (2–3 sentences). [2 × 10 = 20]

- I. Define Artificial Intelligence and mention any two of its real-life applications.
- II. Differentiate between deterministic and probabilistic problems with one example each.
- III. Name FIVE logic gates? Write truth table of ONE.

- IV. Define Machine Learning and Deep Learning in brief.
- V. Mention any two examples each of transactional searches.
- VI. What is a chatbot? Give two examples of widely used chatbots.
- VII. List four social benefits of AI in healthcare.
- VIII. Write any four types of matrices with one example each.
- IX. What is the difference between / and // in Python? Explain with an example.
- X. Write a Python program to swap two variables without using a third variable.

Section C: (Attempt any four questions) [4 × 10 = 40]

Question 3:

- (a) Trace the historical development of Artificial Intelligence and write its major milestones. [5]
- (b) Describe the components of an AI project framework with a neat diagram. [5]

Question 4:

- (a) Explain the applications of ML/DL in image processing, speech recognition, and information retrieval systems. [5]
- (b) Describe the working and advantages of Natural Language Processing (NLP) in AI-based systems like chatbots and translators. [5]

Question 5:

- (a) Explain navigation, informational, transactional, and investigative searches with one example each. [5]
- (b) Explain the potential uses of AI in e-commerce, online marketplaces, and social networking platforms. [5]

Question 6:

- (a) Demonstrate matrix multiplication (2×2 matrices) with an example and explain its uses in AI. [5]
- (b) Explain vector arithmetic (addition, subtraction, dot and cross products) with examples and applications in AI. [5]

Question 7:

- (a) Which method is used to add multiple elements from another list into an existing list? [2]
- (b) Convert the tuple (1, 2, 3, 9, 2, 5) into a set and describe the observed changes. [3]
- (c) Write a Python program to:
 - (a) Create a string “Artificial Intelligence”.
 - (b) Count the number of times “e” appears.
 - (c) Convert the string to uppercase.
 - (d) Replace “Intelligence” with “Learning”.
 - (e) Print the modified string. [5]

Question 8:

- (a) Compare Human Intelligence and Artificial Intelligence in at least five key areas. [5]
- (b) Describe three major types of data: structured, unstructured, and semi-structured with examples. [5]